Introduction

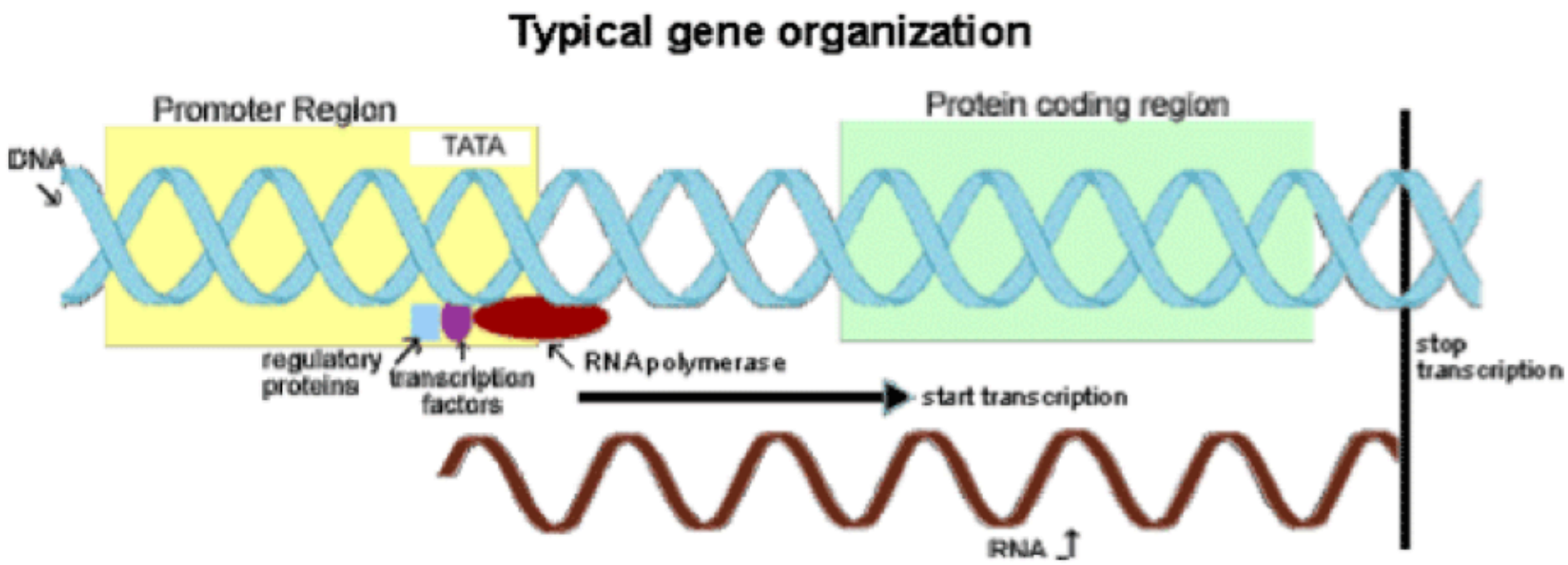
The goal of those two lectures is to use simple statistical analysis techniques on sequences to automatically annotate genomic data. It is a simple extension of the dna-sequence-statistics-2 lecture.

We will annotate two different informations along the genomic sequence:

1. Detect Transcription Factor Binding Sites in promoter sequences
2. Annotate the genes in a genome.

Transcription Factor Binding sites annotation

Below is a reminder about the sequence structure around a gene. The **recruitment** of transcription factor and regulation proteins upstream of the start of the gene direct the transcription of the gene (by the RNA polymerase).



The sequences where the transcription factors bind are called Transcription Factor Binding Sites (TFBS) or promoter sequences. They act basically like switch that can turn the expression of the gene on when activated.

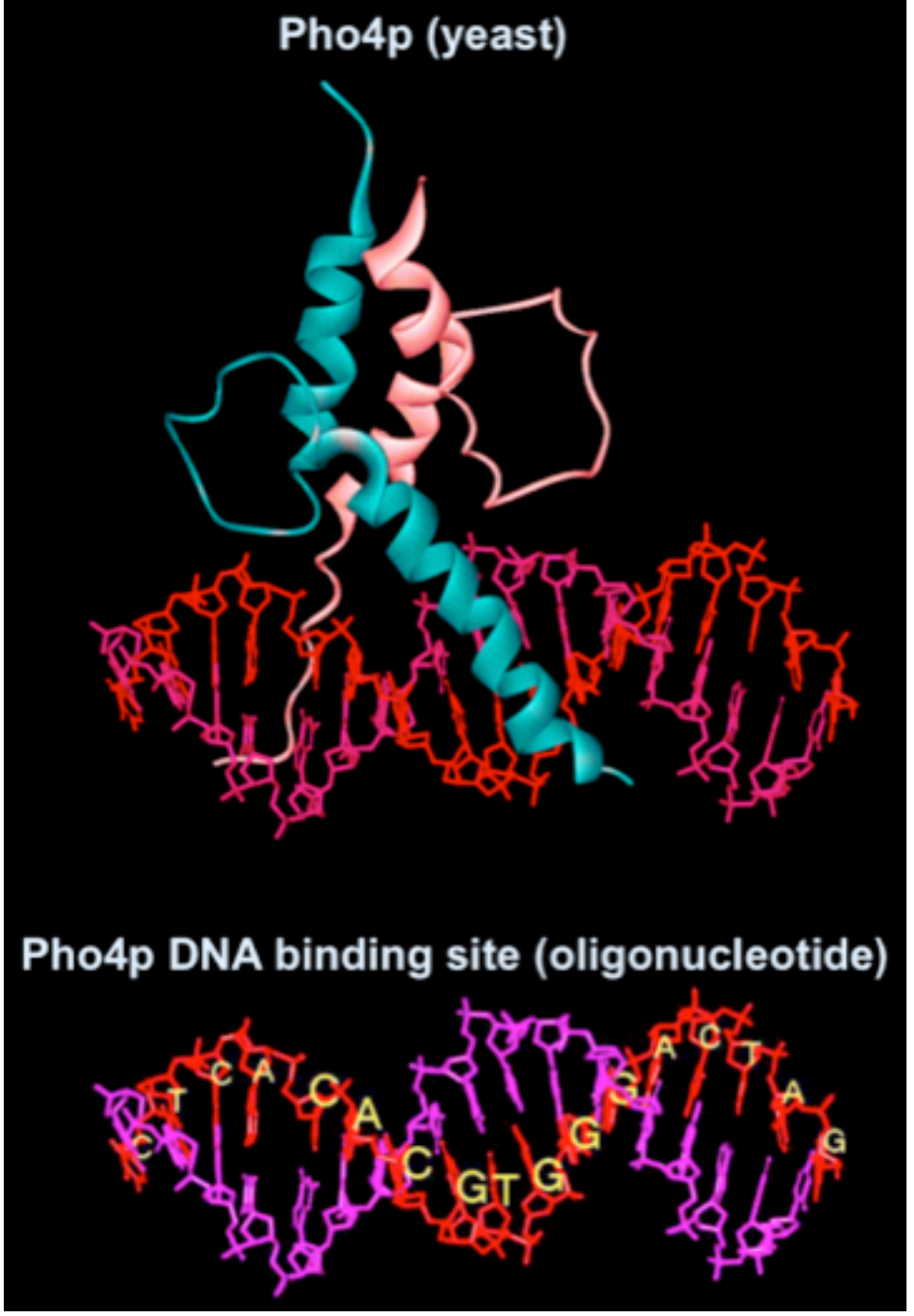


Figure: Protein structure of the transcription factor Pho4p bound to the DNA (top) and sequence of the binding site overlayed on the DNA molecule (bottom)

In general those transcription factors bind to words of length 6 to 10 nucleotides. It is possible to detect for some transcription factor experimentally where it binds, as shown for PHO in the example below.

Gene	Site Name	Sequence	Affinity
PHO5	UASp2	---aCtCaCACACGTGGGACTAGC-	high
PHO84	Site D	---TTTCCAGCACGTGGGGCGGA--	high
PHO81	UAS	----TTATGGCACGTGCGAATAA--	high
PHO8	Proximal	GTGATCGCTGCACGTGGCCCGA---	high
group 1	consensus	---------gCACGTGgg-------	high
PHO5	UASp1	--TAAATTAGCACGTTTTCGC----	medium
PHO84	Site E	----AATACGCACGTTTTTAATCTA	medium
group 2	consensus	--------cgCACGTTtt-------	medium
Degenerate consensus		---------GCACGTKKk-------	high-med

Figure: Example of the TFBS for PHO on two groups of genes. Those sequences are located upstream of the gene sequence, generally between 200 and 500 nucleotide before the transcription start site

As you can see the `GCACGT[TG][TC]` sequence is the binding motif. However performing those experiments is time consuming and costly and cannot be done for each transcription factor.

An alternative way to do is to perform *statistical analysis* on a set of sequence that are coregulated (i.e. transcribed in the same condition). We know those sequences should possess a common binding motif, but we have now to find its sequence. To do so, we will search the sets of k-mers (nucleotide words) that are occurring more than expected given the genomic composition of the words in yeast.

Part 1 Loading sequences and counting k-mers

First we need to estimate the average nucleotide frequency in the genome of the baker's yeast *S. cerevisae*. The sequence of the first four chromosomes can be downloaded here (yeast chr1 to 4) (http://www.lcqb.upmc.fr/hrichard/sequences/yeast_s_cerevisae_genomic_chr1-4.fna)

We put together all the four chromosomes for the next analysis.

In [ ]:

```
library(seqinr)

yeast <- read.fasta(file = "http://www.lcqb.upmc.fr/hrichard/sequences/yeast_s_ce

##Chr1 pos 1 to 100

##let's put all chromosome together
yeastseqs = c()
##Length of the chromosomes
for (i in 1:length(yeast)){
    cat("Chromosome", i, ":", length(yeast[[i]]), "\n")
    yeastseqs = c(yeastseqs, yeast[[i]])
}

cat("total length:", length(yeastseqs) )
##we want to compute the frequency of each nucleotide
```

Let's compute the frequency of the four letters now, that will be handy for later.

In [ ]:

```
nuc.count = table(yeastseqs)

nuc.freq = nuc.count/ sum(nuc.count)

nuc.count

nuc.freq
```

We can remark that there the almost equalities `%a` $\approx$ `%t` and `%c` $\approx$ `%g`. This is a rule due to base pairing and is named Chargaff's pairing rule (https://en.wikipedia.org/wiki/Chargaff%27s_rules).

Part 2 Detect promoter sequences

To detect sequence motif of the PHO binding site, we have a set of sequences (the same as in the figure above) that are all regulated by the PHO transcription factor (PHO sequences) (http://www.lcqb.upmc.fr/hrichard/sequences/regulatory_seq_PHO.fasta).

We will first count the occurrences for all words of a certain size (4 or 6) and compare for each of them the observed count with the expected count, if the sequence has the composition of the *S. cerevisae* genome (the vector `nuc.freq`).

Let's first define the fonction that compute the expected count of a word `w` given letter frequencies `nfreq` and sequence of length `len`.

The probability for a word $w = w_1 \dots w_k$ is simply the product of the probabilities of each letter. For instance $P(TACT) = P(T) \cdot P(A) \cdot P(C) \cdot P(T) \approx 0.308 * 0.309 * 0.191 * 0.308$.

Then if we consider a sequence of length ℓ, there are $\ell - k + 1$ positions where w can appear. So the expected number of occurrences is simply $(\ell - k + 1)P(w)$.

We define the function `expectedFreq` accordingly

In [ ]:

```
expectedFreq <- function(w, nfreq, len){
  eF = 1
  k = length(w)
  for (c in s2c(w)){
    eF = eF * nfreq[c]
    }
  eF = eF*(len - k + 1)
  return(eF)
}
```

Let's read the sequences for PHO:

In [ ]:

```
pho = read.fasta(file = "http://www.lcqb.upmc.fr/hrichard/sequences/regulatory_se

phoseqs = c()
for (i in 1:length(pho)){
    phoseqs = c(phoseqs, pho[[i]])
}

pho.length = length(phoseqs)

cat("Length:", pho.length)
```

We can now compute the number of occurrences for all words of length 4, and compare with the expected count

In [ ]:

```
pho.count4 = count(phoseqs, 4)

##To get the same names for the expected counts
pho.exp4 = pho.count4

for (i in 1:length(pho.exp4)){
  word = names(pho.exp4)[i]
  pho.exp4[i] = expectedFreq(word, nuc.freq, pho.length)
}

#
plot( as.vector(pho.exp4), as.vector(pho.count4),
      pch = 19, col = "blue", cex =0.7,
      xlab = "Expected count", ylab = "Oberved count")
abline(a = 0, b =1, col = "red")
```

The points above the diagonal are observed more often than expected. Let's look how the 25 most overrepresented correspond to the PHO motif.

In [ ]:

```
pho.sig4 = pho.count4 / pho.exp4
##Check the distribution
hist(pho.sig4, br =40)

sorted.sig4 = sort(pho.sig4, decreasing = TRUE)
overrep.words4 = names(sorted.sig4[1:20])

cat(overrep.words4, sep = "\n")
```

First analysis with $k = 4$

- Let's study the 20 most overrepresented words we just produced with $k = 4$. Open them in a notepad, can we align some of them with the PHO motif: `GCACGT[GT][GT][GT]` ?
- are there some words that do not align to the motifs. Can you explain why they are overrepresented in this case?

Extension with $k = 6$

- Redo the analysis with $k = 6$ and get the 20 most overrepresented k mers.
- Try to align those words again with the PHO motif, and put aside words that do not align. Can you construct other motifs?
- Can you propose some adjustments that could improve the results?